Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “kinetic energy” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

kinetic energy: energy associated with motion

scalar: a quantity described by size without a vector direction

01 Warm-up: bring back an earlier idea

An impactor has **3** times the original mass at the same speed and stopping time. **Compare momentum** and average stopping-force size.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar.
Neither alone gives a complete damage forecast.

Worked reasoning: A **2 kg** cart at **3 m/s** has momentum **6 kg·m/s** and kinetic energy $\frac{1}{2}\times 2\times 3^2 = 9$ **J**. A **1 kg** cart at **6 m/s** has the same momentum but **18 J** of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal-mass carts move at speeds **2** and **4 m/s**. **Predict how their momenta** and kinetic energies compare before calculating.

Sentence starter: Both … . They differ in … because … .



Same mass; 2 m/s Same mass; 4 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

The lab carts in question **3** each have mass **2 kg**. One moves at **2 m/s** and the other at **4 m/s**.
1. **Calculate both kinetic energies**.
2. **Draw bars** to **test your prediction about the** effect of doubling speed.

Sentence starter: I predict … . I used … to make this prediction because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

1. **List three forms or destinations** of energy after a deforming collision.
2. Does “less kinetic energy” mean “energy destroyed”?

Sentence starter: I notice … . This supports … because … .

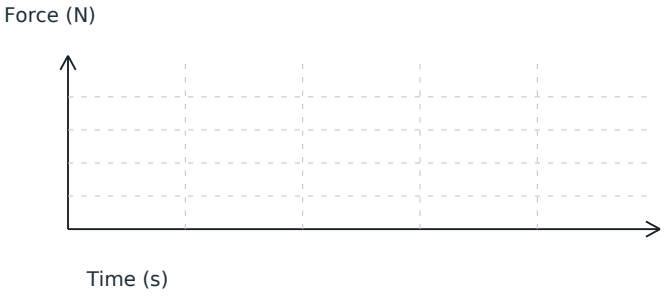
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **24 N** net force acts right for **3 s** on a **4 kg** cart initially at rest. **Find impulse** and final velocity.
Show the signed impulse on the force–time graph.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

07 Exit check

For the same mass, speed triples. By what factor does kinetic energy change? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Find two carts** with equal mass but speeds in a **1**: **2** ratio.
2. **Compare momentum** and energy multipliers.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3